

VIRIDIUM LINUX

Project Documentation

Arch underneath. Performance on top. User in control.

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01

What is Viridium Linux?

Viridium Linux is a performance-focused Linux distribution built on the Arch Linux ecosystem. It provides a streamlined foundation designed for users who want a capable, efficient system while retaining full control over their software and configuration.

Viridium combines the flexibility of Arch with a distinct focus on performance, modern hardware, gaming, and user choice. Rather than replacing the ideas that make Arch successful, Viridium sharpens them: it keeps the transparency and depth of an Arch-based system while smoothing over the parts of setup and maintenance that tend to slow people down.

The project is aimed at a wide range of users — from long-time Arch users who want a faster starting point, to newcomers who want the power of Arch without assembling every layer by hand, to gamers and enthusiasts who care about squeezing the most out of modern hardware. What unites them is a preference for a system that explains itself, stays out of the way, and never locks them into a single way of working.

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The Viridium Philosophy

Viridium is guided by three simple, interlocking ideas. Together they describe both what the project builds and what it deliberately avoids building.

ARCH UNDERNEATH Access to Arch's extensive repositories and established Linux infrastructure.	PERFORMANCE ON TOP Efficient components, modern hardware, gaming workloads, and sensible defaults.	USER IN CONTROL No dictated desktop, applications, or configuration style.
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Arch underneath

Viridium builds upon the Arch ecosystem, providing access to its extensive software repositories and established Linux infrastructure. This means Viridium inherits decades of collective packaging work, documentation, and community knowledge instead of starting from zero.

Performance on top

Performance is a central priority, with attention given to efficient system components, modern hardware, gaming workloads, and sensible defaults. Viridium treats performance as a default expectation rather than an advanced, opt-in configuration step.

User in control

Viridium does not dictate how users should configure their desktop or which applications they should use. Defaults exist to give people a strong starting point, not a ceiling on what the system can become.

03

Built on Arch

Viridium retains the strengths of the Arch ecosystem while providing its own focused identity and system experience. It is not a fork that drifts away from Arch's foundations — it is a distribution that builds a distinct layer on top of them.

- Arch Linux compatibility
- Official Arch repositories
- AUR compatibility
- systemd
- Linux kernel ecosystem
- Broad hardware and software compatibility

Where Viridium diverges from a base Arch install is at the package and build layer: instead of relying on the stock Arch tooling, Viridium ships its own interface — `vbuild` — for resolving, building, and installing software. `vbuild` is designed to remain fully compatible with Arch's package format and repositories, so the depth of the Arch ecosystem stays available even though the day-to-day tooling is Viridium's own.

04

Performance

Performance-focused foundation

Viridium is designed with performance in mind across a broad range of workloads and use cases:

- CPU workloads
- GPU workloads
- Gaming
- System responsiveness
- Modern hardware
- Resource efficiency

This attention to performance runs through the whole stack — from kernel-level scheduling and I/O behavior, to how services start and interact at boot, to the defaults chosen for graphics drivers and storage. The goal is a system that feels responsive under normal, everyday use, not just in synthetic benchmarks.

Sensible optimization

Viridium focuses on practical performance rather than unnecessary system modifications. Optimization should improve the experience without taking control away from the user. Where a default trade-off exists, Viridium prefers the option that is fast, well-tested, and easy to override, over the option that is theoretically fastest but fragile or opaque.

05

User Control

“Your system should remain yours.”

Viridium does not require users to adopt a particular:

- Desktop environment
- Window manager
- Application suite
- Workflow
- Configuration style

Users can build their system around their own needs. Viridium's defaults are meant to be a sensible, working starting point — not a permanent arrangement. Every layer above the base system, from the desktop shell to the shell prompt, is expected to be replaced, reconfigured, or removed entirely without fighting the distribution to do so.

This principle extends to configuration files and system services as well: nothing in Viridium is deliberately hidden or made difficult to inspect. Users who want to understand exactly what their system is doing, and why, are able to do so.

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Software Freedom

Users can choose where their software comes from:

- Arch repositories
- AUR
- Flatpak
- Other compatible software sources

“Viridium provides the foundation. Users decide what they build on top of it.”

All software installation and management in Viridium is handled through vbuild, Viridium's own package and build interface. vbuild is built to speak the same package format and repository structure as the wider Arch ecosystem, so packages from the official Arch repositories and the AUR install cleanly, while Viridium-specific packages and build recipes are handled through the same tool. Flatpak and other sandboxed or cross-distribution formats remain available for users who prefer them, giving people multiple, overlapping ways to get software onto their system.

07

Desktop Freedom

Viridium does not prescribe a desktop environment. The base system is deliberately desktop-agnostic, so the choice of graphical environment is left entirely to the user rather than baked into the distribution's identity.

Users can choose environments and window managers such as:

- KDE Plasma
- GNOME
- Hyprland
- i3
- Sway
- And others

Because Viridium doesn't ship an opinionated desktop layer of its own, switching between a full desktop environment and a minimal tiling window manager is a matter of installing and configuring the components involved, not working around distribution-level assumptions.

08

Gaming

Viridium is designed to be a strong foundation for gaming on Linux, treating gaming as a first-class workload rather than an afterthought.

- Modern hardware support
- Performance-focused kernel
- GPU support
- Steam and other gaming platforms
- Compatibility with the broader Linux gaming ecosystem

This includes attention to the details that matter most for gaming specifically: up-to-date graphics drivers, low-latency kernel behavior, and compatibility with the tools the Linux gaming community already relies on. Because Viridium remains close to upstream Arch, it also benefits quickly from improvements made across the wider Linux gaming ecosystem.

09

Modern Hardware

Viridium is built for modern hardware, with the expectation that most users are running current-generation components rather than legacy systems.

- Modern AMD CPUs
- Modern Intel CPUs
- Modern GPUs
- High-speed storage
- Contemporary networking hardware

Supporting current hardware well means keeping the kernel, drivers, and firmware handling up to date, and making sure new hardware features are usable soon after they become available, rather than months or years later.

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System Foundation

Viridium provides the foundation while leaving the upper layers to the user.

User Software
Desktop / WM / Apps
Viridium Linux
Arch Linux Ecosystem
Linux Kernel
Hardware

Viridium provides the foundation while leaving the upper layers to the user.

At the base sits the hardware itself, followed by the Linux kernel and the wider Arch ecosystem that Viridium draws its packages and infrastructure from. Viridium's own layer — including vbuild and the viridium diagnostic tool — sits above that, providing the tooling that ties the system together. Everything above this point, from the desktop environment to individual applications, is left entirely to the user to choose and configure.

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Viridium Tools

viridium

System management and diagnostics. The `viridium` command is the system's diagnostics and management entry point — it surfaces hardware status, kernel parameters, service state, and Viridium-specific settings in one place, so troubleshooting a system rarely means hunting across multiple unrelated tools.

vbuild

Viridium's package and build interface. `vbuild` is the tool responsible for installing, updating, and removing software on a Viridium system. It resolves dependencies across the official Arch repositories, the AUR, and Viridium's own package sets, builds packages when a prebuilt binary isn't available, and keeps a local cache to make rebuilds and rollbacks fast. Because it speaks Arch's package format natively, existing Arch packaging knowledge and documentation carry over directly to a Viridium system.

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Visual Identity

Colour	Hex
Primary teal	#20C997
Dark teal	#159A78
Light teal	#63E6BE
White	#FFFFFF

Viridium's visual identity uses teal-green accents to represent performance, modern technology, and clarity. The palette is intentionally small and high-contrast, so it reads clearly in both light and dark interface contexts.

Primary teal is used for key actions, highlights, and branding moments. Dark teal is reserved for text, headers, and areas that need stronger contrast, while light teal is used sparingly for secondary accents and hover states.

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Security & Reliability

- Established Arch infrastructure
- Standard Linux security mechanisms
- systemd
- Regular package updates through vbuild
- User-controlled configuration

Security in Viridium is largely inherited from the maturity of the Arch ecosystem it builds on: a well-audited kernel, standard Linux permission and sandboxing mechanisms, and systemd's service isolation features. On top of that, vbuild's dependency resolution and build caching are designed to make it straightforward to keep a system current, so that security-relevant updates can be applied quickly and predictably rather than being deferred.

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Updates

Viridium follows the Arch ecosystem and is designed to receive software and system updates through vbuild, its compatible package and build infrastructure. Because vbuild remains aligned with the Arch package format, updates flow from the same upstream sources that keep the rest of the Arch ecosystem current, rather than from a separate, Viridium-only update channel.

This approach favors a rolling-release model: rather than waiting for periodic, versioned releases, a Viridium system can stay continuously up to date as upstream packages are refreshed, while vbuild's local caching and rollback support give users a way to step back from an update that causes problems.

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Community & Ecosystem

Viridium benefits from the wider Arch and Linux ecosystem:

- Arch documentation
- Arch packages
- AUR
- Linux ecosystem
- Open-source software

Because Viridium stays close to Arch conventions, the enormous body of existing Arch documentation, community troubleshooting knowledge, and package metadata remains directly useful on a Viridium system. Users are not asked to learn an entirely separate ecosystem in order to get help or find software — most of what they already know about Arch, or can learn from Arch's own resources, applies here as well.

Project website: viridiumlinux.github.io

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Project Vision

Viridium aims to provide a focused Linux experience where performance and user control coexist. It is built for users who want a capable foundation without being forced into a particular way of using their computer.

As the project grows, the intention is to keep this balance rather than trade it away: staying close enough to Arch that the ecosystem's strengths remain available, while continuing to invest in the performance-focused tooling — `vbuild` and `viridium` chief among them — that gives Viridium its own identity.

**ARCH UNDERNEATH.
PERFORMANCE ON TOP.
USER IN CONTROL.**

Viridium Linux

Performance-focused. Arch-based. User-controlled.